

# Create synthetic data

Celeste Falcon [email: celestemfalcon@gmail.com]

2026-03-05

```
# Install the synthpop package
# install.packages("synthpop")

# Load the library
library(synthpop)
```

New version of synthpop (1.9-0) with disclosure functions  
see disclosure.pdf for details and NEWS file for other changes

Find out more at <https://www.synthpop.org.uk/>

```
library(readxl)
library(tidyverse)
```

```
-- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
v dplyr      1.2.0      v readr      2.2.0
v forcats    1.0.1      v stringr    1.6.0
v ggplot2    4.0.2      v tibble     3.3.1
v lubridate  1.9.5      v tidyr      1.3.2
v purrr      1.2.1
```

```
-- Conflicts ----- tidyverse_conflicts() --
x dplyr::filter() masks stats::filter()
x dplyr::lag()     masks stats::lag()
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become
```

```
library(broman)
```

```
# Load your original data (replace 'original_data' with the name of your data frame)
# Example with a built-in R dataset:
data(iris)
original_data <- iris

# Generate the synthetic data set
# The syn() function automatically determines the best distribution for each variable
synthetic_data <- syn(original_data)
```

## Synthesis

-----

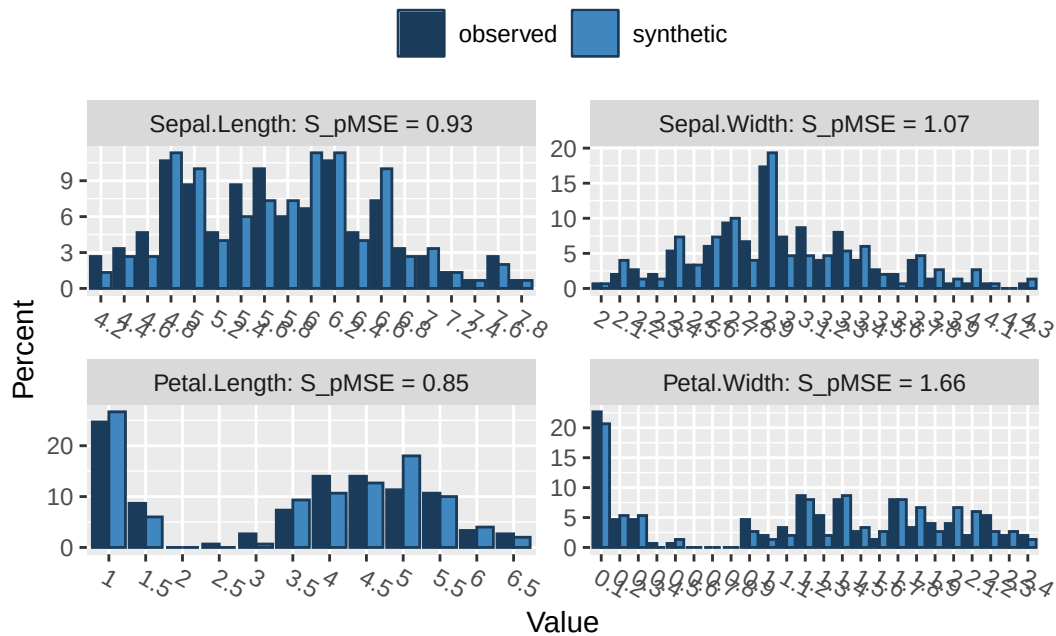
Sepal.Length Sepal.Width Petal.Length Petal.Width Species

```
# Extract the synthetic data frame
synthetic_df <- synthetic_data$syn

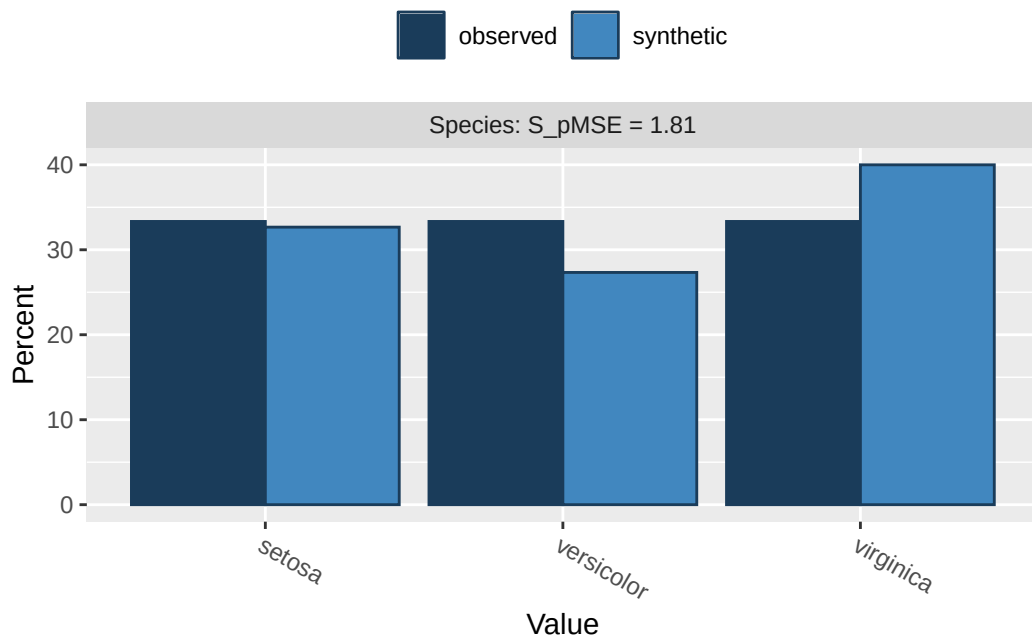
# Compare the statistical properties of the original and synthetic data (optional)
# The package provides tools for this, such as the compare.syn function
compare(synthetic_data, original_data)
```

Calculations done for Sepal.Length  
 Calculations done for Sepal.Width  
 Calculations done for Petal.Length  
 Calculations done for Petal.Width  
 Calculations done for Species

Comparing percentages observed with synthetic



Press return for next variable(s):



Selected utility measures:

|              | pMSE     | S_pMSE   | df |
|--------------|----------|----------|----|
| Sepal.Length | 0.001548 | 0.928538 | 4  |
| Sepal.Width  | 0.001791 | 1.074428 | 4  |
| Petal.Length | 0.001418 | 0.851037 | 4  |
| Petal.Width  | 0.002768 | 1.661019 | 4  |
| Species      | 0.001508 | 1.809302 | 2  |

```
bioassay_data <- read_xlsx("./input/anonymized_for_portfolio/Bioassay Results_anonymized.xlsx")
  filter(!is.na(Dead)) %>%
  mutate(Week = as.factor(Week)) %>%
  mutate(Sample = as.factor(Sample)) %>%
  mutate(Replicate = as.factor(Replicate)) %>%
  mutate(Duration = as.factor(Duration))
```

```
bioassay_data_control <- bioassay_data %>%
  filter(Sample == "CONTROL")
```

```
bioassay_data_treatment <- bioassay_data %>%
  filter(Sample == "TREATMENT")
```

```
# # full data set
# bioassay_data_syn <- syn(bioassay_data, visit.sequence = c("Dead", "Total", "Mortality"))
# bioassay_data_syn_df <- bioassay_data_syn$syn
# compare(bioassay_data_syn, bioassay_data, vars = c("Mortality"), plot = F)

# control data set
bioassay_data_control_syn <- syn(bioassay_data_control, visit.sequence = c("Dead", "Total", "Mortality"))
```

CAUTION: Your data set has fewer observations (72) than we advise.

We suggest that there should be at least 130 observations

(100 + 10 \* no. of variables used in modelling the data).

Please check your synthetic data carefully with functions

compare(), utility.tab(), and utility.gen().

Variable(s): Total numeric but with only 1 or fewer distinct values turned into factor(s) for

Variable Total has only one value so its method has been changed to "constant".

Variable Total removed as predictor because only one value.

Variables Dead, Mortality are collinear. Variables later in 'visit.sequence'

are derived from Dead.

Variable(s): Week, Sample, Replicate, Duration not synthesised or used in prediction.  
CAUTION: The synthesised data will contain the variable(s) unchanged.

## Synthesis

-----

Dead Total Mortality

```
bioassay_data_control_syn_df <- bioassay_data_control_syn$syn
# compare(bioassay_data_control_syn, bioassay_data_control, vars = c("Mortality"), plot = T)
compare(bioassay_data_control_syn, bioassay_data_control, vars = c("Mortality"), plot = F)
```

Calculations done for Mortality

Selected utility measures:

|           | pMSE     | S_pMSE   | df |
|-----------|----------|----------|----|
| Mortality | 0.001241 | 0.714703 | 2  |

```
# treatment data set
bioassay_data_treatment_syn <- syn(bioassay_data_treatment, visit.sequence = c("Dead", "Total"))
```

CAUTION: Your data set has fewer observations (72) than we advise.  
We suggest that there should be at least 130 observations  
(100 + 10 \* no. of variables used in modelling the data).  
Please check your synthetic data carefully with functions  
compare(), utility.tab(), and utility.gen().

Variable(s): Total numeric but with only 1 or fewer distinct values turned into factor(s) for

Variable Total has only one value so its method has been changed to "constant".  
Variable Total removed as predictor because only one value.  
Variables Dead, Mortality are collinear. Variables later in 'visit.sequence'  
are derived from Dead.

Variable(s): Week, Sample, Replicate, Duration not synthesised or used in prediction.  
CAUTION: The synthesised data will contain the variable(s) unchanged.

## Synthesis

-----

Dead Total Mortality

```
bioassay_data_treatment_syn_df <- bioassay_data_treatment_syn$syn
compare(bioassay_data_treatment_syn, bioassay_data_treatment, vars = c("Mortality"), plot = 1)
```

Calculations done for Mortality

Selected utility measures:

|           | pMSE     | S_pMSE   | df |
|-----------|----------|----------|----|
| Mortality | 0.000574 | 0.330909 | 2  |

Distribution plots not shown to protect client's real data.

pMSE is propensity mean squared error, a measure of the utility of synthetic data based on how distinguishable it is from the original, real data. Values closer to 0 indicate high utility (the synthetic data is indistinguishable from the real data).

S\_pMSE is its standardized ratio to account for scale of measurement for each variable

```
# write.csv(bioassay_data_syn_df, "./synthetic_data.csv", row.names = F)

bioassay_data_ctrl_trt_syn <- bind_rows(bioassay_data_control_syn_df, bioassay_data_treatment_syn_df)

# write.csv(bioassay_data_ctrl_trt_syn, "./synthetic_data_by_sample.csv", row.names = F)
```